Liwei Deng

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Education

College of Electronic and Information Engineering, Shenzhen University ShenZhen, China
Bachelor of Electronic Information Engineering *September,2021-July,2025*

- GPA: **4.02**/4.5 (**89.3**/100); Rank: **19** of Department(19/217)
- **Relevant Courses**: Calculus A(4.5/4.5), Linear Algebra(4.5/4.5), Introduction to Data Mining(4.5/4.5), Theory of Deep Learning(4.0/4.5), Graph Deep Neural Networks: Theories and Practice(4.0/4.5), Deep learning framework setup based on Python(4.0/4.5)

Publications

[1] **Liwei Deng**, Hao Wang* and et al., "Patch Attention Based Adaptive Multihead Graph Convolutional Networks for Traffic Flow Forecasting" submitted to *Neurocomputing* (**JCR Q1**).

[2] Xiaobao Song, Hao Wang*, **Liwei Deng**, et al., "D2Vformer: A Flexible Time Series Prediction Model Based on Time Position Embedding" has been published on Arxiv, doi: https://doi.org/10.48550/arXiv.2409.11024, and is being prepared for submission to *IEEE Transactions on Intelligent Transportation Systems(ITS)* (**CAS Q1 Top, JCR Q1**).

[3] **Liwei Deng**, Hao Wang*, et al., "STD2Vformer: A Flexible Spatial-Temporal Model Based on Temporal and Spatial Embedding," submitted to *IEEE Transactions on Industrial Informatics(TII)* (**CAS Q1 Top, JCR Q1**).

[4] Xiaobao Song, **Liwei Deng**, Hao Wang* and et al., "Deep Learning-based Time Series Forecasting", **acceped by** *Artificial Intelligence Review* (**IF=12.0, JCR Q1**).

[5] Xiaobao Song, **Liwei Deng**, Hao Wang*, Yaoan Zhang, Zuosheng Chen, Wenming Cao and Yuxin He, "Long-term Time Series Prediction Model Based on Deep Learning and Temporal Decomposition", **accepted by** *Computer Engineering and Applications*.

Patents

[a] Hao Wang, Xiaobao Song, Zuosheng Chen, **Liwei Deng** and Yaoan Zhang, "The Deep Learning-based Power Prediction Method, Device, and Storage Medium (基于深度学习的电力预测方法、装置及存储介质)" , Chinese Patent for Invention, No. CN202310687810.8, filed.

Skills

English: **IELTS: 6.5, GRE: 336**

Proficient in: Pytorch, Python, LaTeX

Mastery of: Linux, Office 365, Matlab, Git

Research Experience

Guangdong Multimedia Information Service Engineering Technology Research Center SZU
Research Assistant(Supervised by Assistant Professor Hao Wang) 09/2022-Present

An Algorithmic Study of Traffic State Prediction Problem Based Deep Learning[1] 07/2023-Present

- We reproduced over 25 traffic state prediction models and found that existing models fail to effectively capture different traffic patterns at different nodes, resulting in low prediction accuracy.
- We introduced PA^2MGCN, an innovative spatio-temporal graph neural network specifically designed for traffic prediction.

- PA^{2}MGCN stands out by effectively capturing fine-grained node patterns by integrating the Adaptive Multihead GCN and the Patch Temporal Attention module, which we propose.

An Algorithmic Study of Time Embedding Method Based Deep Learning[2],[3] 07/2023-Present

- For time series and spatial-temporal data, traditional models cannot achieve flexible prediction.
- Therefore, for time series, we introduced an innovative time position embedding method known as Date2Vec (D2V) and presented a novel time series prediction model called D2Vformer, which can generate temporally flexible predictions.
- Similarly, for spatial-temporal data, we introduced an innovative temporal and spatial embedding method known as STD2V and presented a novel spatial-temporal prediction model called STD2Vformer, which can generate both temporally and spatially flexible predictions.
- D2Vformer and STD2Vformer can accurately predict variable-length sequences that are highly relevant in real-world scenarios.

A Survey: Time Series Forecasting Model From 2014 To 2024[4] 10/2022-05/2024

- Due to the lack of a guide in the field of time series forecasting, which can assist non-experts or beginners in quickly deploying suitable time-series prediction models for their tasks.
- We reproduced over 30 time-series prediction models from 2014 to 2024, summarizing the innovations of each model and highlighting the shortcomings of current models.
- We have also developed a methodological guide for time-series prediction tasks by conducting extensive experiments on eight real-world datasets.

An Algorithmic Study of Long-Term Time Series Prediction Model[5],[a] 10/2022-07/2023

- For time-series forecasting, current models fail to effectively capture long-term seasonal dependencies, making it difficult to produce accurate long-term predictions.
- By integrating the Attention mechanism with temporal decomposition, we introduced a novel Trend Attention within patches and Season Attention between patches.
- This resulted in a new long-term time-series prediction model called DeepTD-LSP, which effectively captures both short-term trends and long-term seasonal components to produce accurate long-term predictions.

Summary: Through participating in research projects, I have developed a solid understanding of time-series and spatio-temporal prediction models, gaining insights into the existing issues in the field and potential future research directions.

Honor & Award

Honor

- Third Prize for Outstanding Academic Achievement November, 2022
- Third Prize for Outstanding Academic Achievement November, 2023

Competition Award

- **Second Prize** in the 2023 Mathorcup University Mathematical Modeling Challenge June, 2023
- **Second Prize** in the 2023 China Undergraduate Mathematical Contest in Modeling (CUMCM) in Guangdong Province January, 2024
- Third Prize in the 15th National Professional Software Engineering 'Blue Bridge Cup' Design Contest in Guangdong Province May, 2024